High-Performance Data Engineering and Distributed Sharding

Relational and NoSQL Databases

Modern data engineering pipelines process high-velocity transactions and analytical queries. We review the characteristics of relational PostgreSQL databases and NoSQL document stores.

Database Sharding and Partitioning

To scale write throughput, we implement distributed database sharding. Table records are partitioned across active nodes based on shard keys. Relational indexes and execution plan optimizations ensure low query latency.

ETL Workflows and Stream Processing

We design robust ETL data pipeline workflows using Apache Kafka and Flink. Raw events are streamed in real-time, aggregated, and loaded into a columnar data warehouse for analytics queries.

Summary of ACID and Consistency

Maintaining database transactional consistency requires strict ACID compliance. Our distributed write-ahead logs (WAL) ensure replica replication is fully fault-tolerant during concurrent updates.